EEE303 Tutorial 3

1. A DC generator is connected to a 220 V DC main (Figure T3-1). The current delivered by the generator to the mains is 100 A. The armature resistance is 0.1 Ω. The generator is driven at a speed of 400 rpm. Assume constant flux, calculate
   (a) the induced emf
   (b) the electromagnetic torque
   (c) the mechanical power input to the armature neglecting iron, windage and friction losses

   (d) the power input and output of the armature when the speed drops to 350 rpm. State whether the machine is generating or motoring.

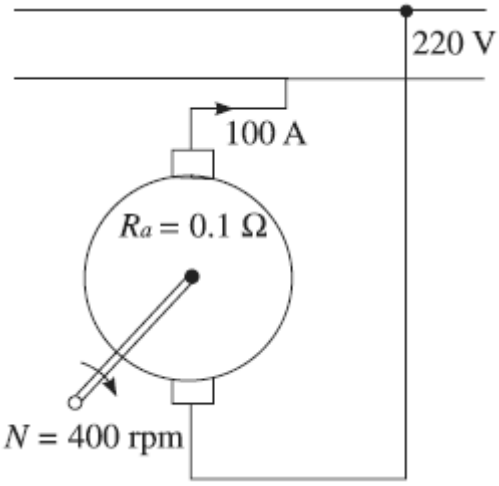
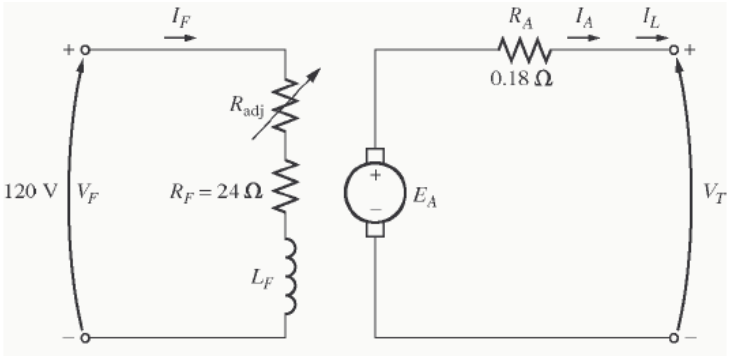


Figure T3-1

2. A shunt generator delivers 50 kW at 250 V and 400 rpm. The armature resistance is 0.02 Ω and field resistance is 50 Ω. Calculate the speed of the machine when running as a shunt motor and taking 50 kW input at 250 V.
3. A DC shunt machine, connected to 250 mains, has an armature resistance of 0.12 Ω, and the resistance of the field circuit is 100 Ω. Calculate the ratio of the speed as a generator to the speed as a motor, the line current in each case being 80 A.
4. A separately excited generator is rated at 6 kW, 120 V, 50 A, and 1800 r/min and its field circuit is rated at 5A. Other parameters about the machine are shown in Figure T3-2a. The magnetization curve for a separately excited dc generator is shown in Figure T3-2b, assuming no armature reaction.



$R_A = 0.18\ \Omega$ $V_F = 120$ V

$R_{adj} = 0$ to $30\ \Omega$ $R_F = 24\ \Omega$

$N_F = 1000$ turns per pole

Figure T3-2a

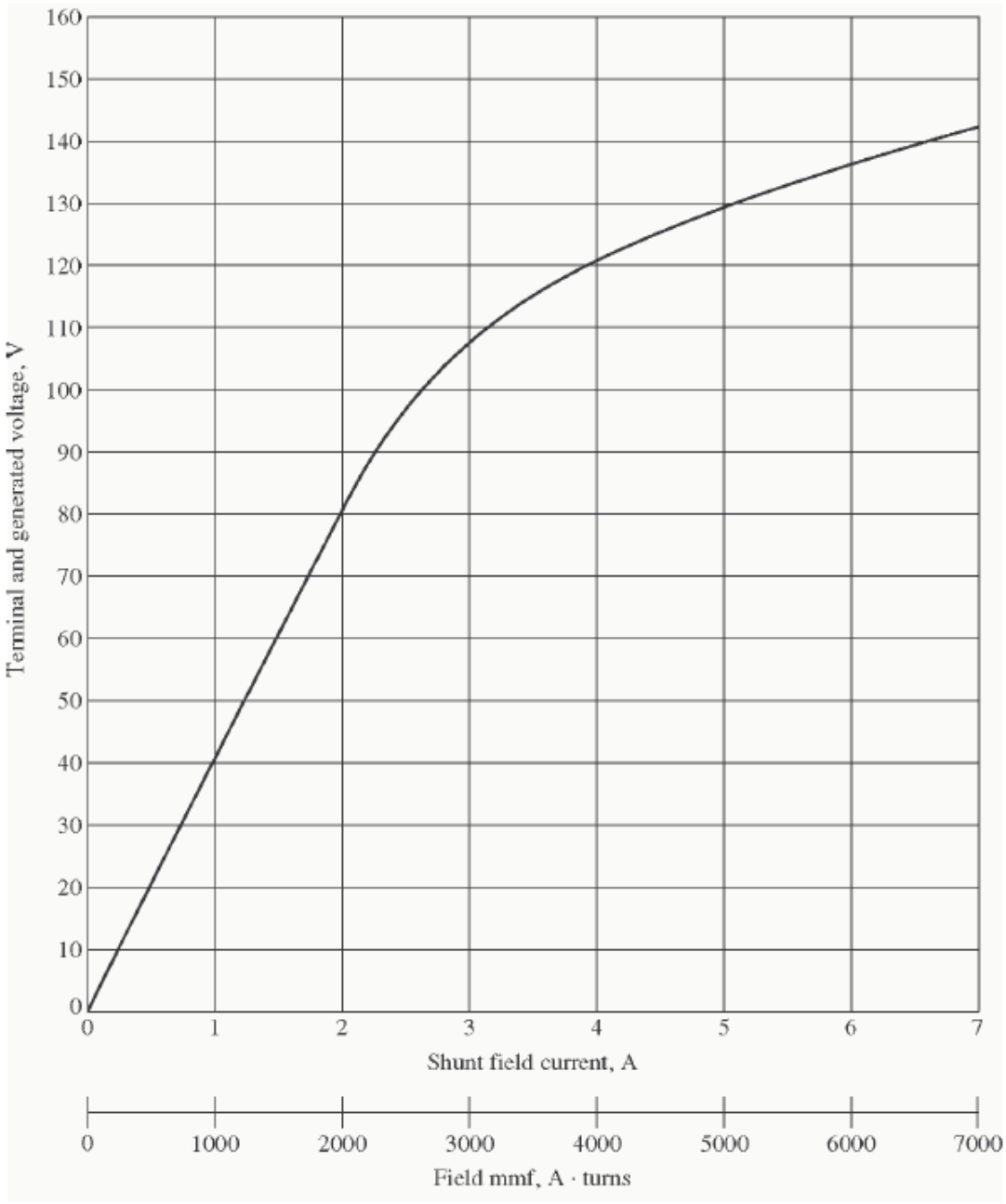


Figure T3-2b

(a) If this generator is operating at no load, what is the range of voltage adjustments that can be achieved by changing R_{adj}?

(b) If the field rheostat is allowed to vary from 0 to 30 Ω and the generator's speed is allowed to vary from 1500 to 2000 r/min, what are the maximum and minimum no-load voltages in the generator?

(c) If the armature current of the generator is 50 A, the speed of the generator is 1700 r/min, and the terminal voltage is 106 V, how much field current must be flowing in the generator?

(d) Taking armature reaction into consideration, at full load equivalent to 400 A·turns of magnetomotive force, find the terminal voltage of the generator when I_F = 5 A, n_m = 1700 rpm, and I_A = 50 A?

5. A separately excited DC generator turning at 1400 rpm produces a generated voltage of 127 V under no load conditions. The armature resistance is 2 Ω. If the machine then delivers a current of 12 A into a load, calculate
   (a) the terminal voltage
   (b) the power dissipated in the armature winding
   (c) the braking toque exerted by the generator on the prime mover
6. A 240 kW, 500 V, 1750 rpm separately excited DC generator has an overall efficiency of 94%. The shunt field winding has a resistance of 60 Ω, and the rated field winding current is 5 A. the resistive loss in the armature winding is 5.52 kW at full rated current. Calculate
   (a) the rated armature current
   (b) copper and rotational losses in the machine at full rated current
   (c) resistive losses in the armature when the machine is operating at half rated current

   (d) the torque required from the prime mover to drive the generator at 1750 rpm at half rated current, assume generator still supplies field losses despite being separately excited.